GroupTrack

V3.0 Implementation Plan — Final

March 29, 2026 | Branch: feature/convoy-event-ride | Gate: Wednesday Field Test

1. Executive Summary

GroupTrack V3.0 migrates from a fully offline standalone app to a cloud-assisted platform while preserving every V2.4 capability unchanged. The migration is built in two isolated phases with a hard Wednesday field test gate between them.

Core principle: the radio write engine never changes. The only V3.0 change to the radio flow is where the ride JSON originates — downloaded from RDS instead of read from a local file. WorkingConfig assembly, binary build, and BLE write are identical to V2.4.

The schema is built complete from day one — all 12 tables including V4.0 organization structure defined and indexed but never populated in V3.0. All org FKs are nullable. Nothing in V3.0 ever sets them.

2. Intellectual Property Boundary

This boundary must be understood and respected throughout development. It protects GroupTrack IP and defines the open source attribution requirements.

GROUPTRACK LAYER — YOUR IP	MESHTASTIC LAYER — OPEN SOURCE
Ride definition → RDS (name, date, channel, PSK)	**InstallProfileUseCase — accepts binary, writes to radio**
master_config.json — 47-field baseline model	**BLE protocol implementation**
Merge logic — WorkingConfig assembly	**Radio firmware**
ConvoyProfileBuilder — serializes to DeviceProfile binary	**DeviceProfile protobuf structure**
The binary itself — your data, your structure, your output	The binary executor — open source just writes bytes
Hosted ride management, API, RDS schema	Nothing above this line

★ GroupTrack is not a Meshtastic extension. It is an independent product that uses Meshtastic as a radio driver.

★ Radio programming manipulates master config and ride components — this is GroupTrack technology. Open source just implements the binary.

★ The convoy/ directory, ConvoyEngine, hosted API, and RDS schema are original works. GPL does not require sharing new files added to a fork — only modifications to existing Meshtastic files.

3. Open Source Attribution Requirements

GroupTrack uses open source components that require attribution. These disclosure requirements must be implemented at specific points in the V3.0 build. Failure to include them before Google Play submission is a legal risk.

3.1 Components Requiring Attribution

COMPONENT	LICENSE	REQUIREMENT	NOTES
Meshtastic-Android	GPL v3	Attribution + source availability	Modified source must be available on request. New files (convoy/) are not modifications.
Leaflet	BSD 2-Clause	Copyright notice	Attribution control already visible on map. Do not suppress it.
Esri Tile Sources (SAT/TOPO)	Esri ToS	Attribution text on map	Free for non-commercial and small commercial. Verify as scale grows.
Carto (RD tiles)	Carto ToS	© OpenStreetMap + © CARTO	Attribution required in map view.
Google HYB tiles	**Google ToS — VIOLATION RISK**	**REPLACE BEFORE V3.0**	**Replace with Esri overlay. Task zero Monday morning before any AWS work.**

3.2 Attribution Disclosure Points — Required in V3.0

#	LOCATION	CONTENT	TRIGGER	PHASE
1	**TASK ZERO — Replace Google HYB**	**Replace Google tile URL with Esri Transportation overlay in ConvoyConfig.kt**	**Monday morning**	**Before AWS**
2	Leaflet attribution control	Dynamic text per tile source. SAT: © Esri. RD: © OpenStreetMap, © CARTO. Updates when user switches source.	Map settings open	Phase 1
3	First launch disclosure screen	One-time screen: GroupTrack uses open source components. Tap to review. ACCEPT button. Shown once, cached in SharedPreferences.	First run only	Phase 2
4	About Screen	Full attribution block: Meshtastic GPL v3, Leaflet BSD 2-Clause, Esri, OpenStreetMap, CARTO. Links to each. Settings → About.	Settings tap	Phase 2
5	Google Play Store listing	Acknowledgements section: Meshtastic, Leaflet, Esri, CARTO, OpenStreetMap.	Before submission	Pre-launch
6	GroupTrack website	Legal/Attribution page linked from footer. grouptrack.org/attribution	Permanent	Phase 2

7	APK / GitHub repo	LICENSE.txt (GPL v3) and ATTRIBUTION.txt in repo root.	Source dist.	Phase 1

3.3 Package Rename — Required Before Google Play

Current package: com.geeksville.mesh.google.debug. Required for Google Play: com.grouptrack.android

The rename is the formal declaration that GroupTrack is its own product. Beta testers on the old package name retain their working standalone app. New users through Google Play get the V3.0 hosted flow. No forced migration. No shared data conflicts between old and new installs.

> ⚠️ **Add IS_STANDALONE_BUILD = true to ConvoyConfig.kt in Phase 1. When building Google Play version this flag flips to false and standalone ride creation UI is conditionally hidden. One flag, clean separation, no duplicate codebases.**

4. Complete Database Schema — V3.0 Final

All 12 tables created in a single SQL file: create_schema_v3_final.sql. Run against RDS before any Android work begins.

4.1 Table Inventory

TABLE	VERSION	V3.0 STATUS	KEY DESIGN DECISION
organizations	V4.0	**DEFINED EMPTY**	Defined now so no schema migration needed in V4.0. Never populated in V3.0.
users	V3.0	**ACTIVE**	primary_org_id nullable FK — never set in V3.0. email_opt_in controls broadcast eligibility.
kml_routes	V3.0	**ACTIVE**	file_hash UNIQUE — MD5 computed on device. Duplicate silently discarded. source_ride_id tracks provenance.
rides	V3.0	**ACTIVE**	org_id nullable (V4.0). route_id nullable (one KML max). broadcast_sent prevents duplicate email shots.
invites	V3.0	**ACTIVE**	Private rides: invite token is the only access path. Public rides: token used for enrollment tracking.
enrollments	V3.0	**ACTIVE**	Rider accepts invite. callsign and vehicle_type captured for ride roster display.
org_members	V4.0	**DEFINED EMPTY**	UNIQUE KEY uq_org_user prevents duplicate memberships. Never populated in V3.0.
follows	V3.0	**ACTIVE**	User → user only in V3.0. UNIQUE KEY prevents duplicate follows. Broadcast query: WHERE following_id = organizer AND status = active.
notifications	V3.0	**ACTIVE**	Check before insert: COUNT(*) WHERE user_id AND ride_id AND type = broadcast. If > 0 skip — already notified.
email_templates	V3.0	**ACTIVE — 1 seed**	org_id NULL = GroupTrack branding. SET = org override in V4.0. Swap branding without code deploy.
email_queue	V3.0	**ACTIVE**	EC2 cron processes pending rows. Max 3 retries. html_body pre-rendered at queue time captures branding at send moment.
ride_surveys	V3.0	**ACTIVE**	UNIQUE KEY uq_survey — one per rider per ride. KML donation processed in same POST /ride/survey call.

4.2 Critical Schema Design Decisions

Circular FK — rides and kml_routes

rides.route_id references kml_routes. kml_routes.source_ride_id references rides. Both nullable. No circular insert issue because rides are created first with route_id = NULL. KML is donated post-ride with source_ride_id set. The ALTER TABLE statement in the SQL file adds the kml_routes FK after the rides table exists.

KML Duplicate Prevention

MD5 hash computed on device before any upload attempt. API receives hash first, checks kml_routes.file_hash UNIQUE column. If exists: return { accepted: false } — no error message to rider, no file transfer, no server load. Survey continues silently. Original donor attribution is permanent and never threatened.

If hash does not exist: accept file, store to EC2 at /var/www/html/uploads/kml/{route_id}.kml, insert kml_routes row, return { accepted: true, route_id: uuid }.

Ride Visibility Model

RIDE TYPE	org_id	is_public	ACCESS + DISTRIBUTION
Personal private	NULL	FALSE	Invite token only. No broadcast. No discovery.
Personal public	NULL	TRUE	Invite token + broadcast to followers + email shot.
Org ride default	SET	TRUE	Public by default. Org members + followers + email shot. Application layer sets TRUE when org_id set.
Org ride private override	SET	FALSE	Invite only despite org affiliation. Organizer overrides at creation.

Post-Ride Survey and KML Donation Flow

Triggered when rider taps STOP RECORD. One POST /ride/survey call handles both survey and optional KML donation atomically. The kml_donation field in the JSON payload is optional — absent if rider skips upload.

KML is named by the rider at donation time in the post-ride survey. The route name, description, zip, and state are all supplied by the rider in that moment — not inferred from the ride record.

If KML already exists (hash match): silently discard. No message. No rejection dialog. Survey completes identically. The rider never knows the outcome.

5. Complete API Reference

METHOD	ENDPOINT	AUTH	DESCRIPTION
POST	/users/register	Google ID token	Create or update user. Returns user_id UUID. Cached in SharedPreferences on device.
POST	/rides	user_id	Create ride. Body: ride_name, channel_name, channel_psk, ride_date, is_public, map_bounds*, route_id optional. Returns ride_id.
POST	/rides/{id}/invite	user_id (organizer)	Generate invite token. Returns invite URL: http://grouptrack.org/invite/{token}. Opens Android share sheet.
GET	/ride/{token}	None — token is auth	Rider downloads ride JSON via invite link. Returns full ride record including map_bounds, route KML URL if set.
POST	/enroll	user_id	Rider enrolls in ride. Body: ride_id, callsign, vehicle_type.
POST	/ride/survey	user_id	Post-ride survey + optional KML donation. Body: ride_id, rating, notes, kml_donation (optional). Atomic — both processed in one call.
POST	/kml/donate	user_id	Called internally by /ride/survey. Checks file_hash — if exists returns {accepted:false} silently. If new: stores file, inserts row, returns {accepted:true,route_id}.
POST	/broadcast/{ride_id}	user_id (organizer)	Triggers email shot to all followers. Checks broadcast_sent flag — if TRUE returns 409. Queries follows, inserts email_queue rows, sets broadcast_sent=TRUE.
POST	/follows/{organizer_id}	user_id	Rider follows an organizer. UNIQUE KEY prevents duplicates at DB level.
DELETE	/follows/{organizer_id}	user_id	Unfollow. Also deletes pending email_queue rows for this user.
GET	/rides	user_id	List organizer's rides. V3.x.

6. Phase 1 — Isolated Build (Safe Now, No Device Risk)

6.1 Task Zero — Replace Google HYB Tiles (Monday Morning First)

ITEM	DETAIL
What	Replace Google HYB tile URL with Esri World Transportation overlay in ConvoyConfig.kt. One line change.
New URL	https://server.arcgisonline.com/ArcGIS/rest/services/Reference/World_Transportation/MapServer/tile/{z}/{y}/{x}
Why	Google Maps tile URLs require a Maps Platform API key and billing account. Distributing without one violates Google ToS — a real legal risk before Google Play submission.
Result	Esri Transportation overlay on SAT base = same satellite + roads visual. No new dependency. Already using Esri for SAT and TOPO.
Risk	**NONE — one config line**
Est.	15 minutes including build and deploy

6.2 Backend Tasks

#	ID	Task	Risk	Est.
1	**B-01**	Run create_schema_v3_final.sql against RDS. Creates all 12 tables, indexes, and email template seed row.	**NONE**	30 min
2	**B-02**	Verify schema: SHOW TABLES (expect 12). SELECT template_name FROM email_templates (expect 1 row). DESCRIBE kml_routes (confirm file_hash UNIQUE).	**NONE**	15 min
3	**B-03**	Create /var/www/html/uploads/kml/ directory on EC2. Set permissions: sudo chmod 755 uploads/kml/	**NONE**	5 min
4	**B-04**	Test POST /users/register with curl — confirm RDS row created. Test POST /rides — confirm ride_id returned.	**NONE**	20 min

6.3 Android Tasks — New Files Only

#	ID	Task	Risk	Est.
5	**A-01**	Add API_BASE_URL = "http://34.224.89.217/" and IS_STANDALONE_BUILD = true to ConvoyConfig.kt.	**NONE**	5 min
6	**A-02**	Add LICENSE.txt (GPL v3) and ATTRIBUTION.txt to repo root. Commit.	**NONE**	15 min
7	**A-03**	Add dynamic Leaflet attribution text per tile source in convoy_map.html. Updates when user switches SAT/HYB/TOPO/RD.	**LOW**	30 min
8	**A-04**	Add Google Sign-In dependency to build.gradle: com.google.android.gms:play-services-auth:21.x.x	**NONE**	10 min
9	**A-05**	Create ConvoyApiClient.kt — OkHttp wrapper for all API	**NONE**	90 min

		endpoints. Suspend functions. Returns Result<T>. New file only — not imported by anything yet.		
10	**A-06**	Build ConvoySignInScreen.kt — Google One Tap. On success: POST /users/register, cache user_id in SharedPreferences. NOT wired to nav graph yet.	**NONE**	2.5 hrs
11	**A-07**	Add invite deep link handler — AndroidManifest intent-filter for grouptrack.org/invite/ + ConvoyInviteHandler.kt. GET /ride/{token} → save to convoy_events/. If user_id missing: queue download, prompt sign-in first.	**LOW**	90 min
12	**A-08**	Build post-ride survey screen — triggered on STOP RECORD. Rating, notes, optional KML upload. MD5 hash computed on device. POST /ride/survey. kml_donation field absent if skipped.	**LOW**	2 hrs
13	**A-09**	Lead Lock UI buttons — SET AS LEAD (NODE panel), I AM LEAD + RECALC LEAD (MY CART panel). Logic committed — UI only.	**LOW**	90 min
14	**A-10**	Build and install Phase 1 build. Full V2.4 regression test — radio write, offline maps, lead lock, auto-pan. Confirm nothing broken before Wednesday.	**LOW**	30 min

7. Wednesday Gate — V2.4 Field Test

⚠️ Phase 2 does not begin until all tests below pass. This gate protects the proven offline radio engine.

#	ID	Test	Status
1	**FT-01**	Lead lock — drive 1/4 mile after RECORD. Lead locks, single clean track on switchbacks.	**PENDING**
2	**FT-01B**	SET AS LEAD, I AM LEAD, RECALC LEAD — all three overrides verified working.	**PENDING**
3	**FT-04**	Offline maps render cleanly at z18 across SAT area. No grey patches.	**PENDING**
4	**FT-05**	Maps full screen at all zoom levels. No half-screen or spotty tiles.	**PENDING**
5	**FT-06**	No snap-back on zoom, pan, or search. GROUP/MY CART re-centers correctly.	**PENDING**
6	**FT-02**	Channel utilization 23-25% at 5 nodes.	**VALIDATED**
7	**FT-03**	BT stability with Unrestricted battery. No timeout.	**DOCUMENTED**

8. Phase 2 — Integration Wiring (Post-Wednesday Only)

#	ID	Task	Connection Point	Risk	Est.
15	**W-01**	Wire sign-in into app launch — check SharedPreferences for user_id. If missing: navigate to ConvoySignInScreen. If offline or fails: proceed to map with banner.	MainActivity / nav graph	**MED**	1 hr
16	**W-02**	Test sign-in E2E — sign in → POST /users/register → RDS row → user_id cached → app proceeds.	Full flow test	**MED**	30 min
17	**W-03**	Wire ride creation to POST /rides — replace local-only save with API call + local cache write. Add offline fallback: if API fails, save local only with sync-pending flag.	ConvoyViewModel.createRide()	**MED**	90 min
18	**W-04**	Wire invite generation — SEND INVITE button → POST /rides/{id}/invite → share sheet with invite URL.	Ride detail screen	**LOW**	1 hr
19	**W-05**	Test invite flow E2E — organizer sends invite → rider taps link → ride downloaded → radio write succeeds.	Full flow test	**MED**	1 hr
20	**W-06**	Wire automated map download on invite accept — check map_bounds in ride JSON → trigger ConvoyTileDownloader. KML overlay loaded from route_id if set.	ConvoyInviteHandler	**LOW**	90 min
21	**W-07**	Wire broadcast — organizer taps BROADCAST on public ride → POST /broadcast/{ride_id} → email_queue populated → EC2 cron sends.	Ride detail screen	**LOW**	1 hr
22	**W-08**	Build About screen — full attribution block with links. Settings → About. First-launch disclosure screen — one-time, cached in SharedPreferences.	New screen	**LOW**	1 hr
23	**W-09**	Add attribution page to grouptrack.org. Update Google Play store listing acknowledgments section.	Website + store	**LOW**	1 hr
24	**W-10**	Full V3.0 E2E test — sign in, create ride, send invite, rider accepts, tiles auto-download, KML overlays, radio write, mesh operation, post-ride survey, KML donation.	Full system	**MED**	2 hrs

9. Separation Rules — Must Never Be Crossed Before Wednesday

RULE	RATIONALE
ConvoySignInScreen NOT in nav graph until W-01	Unreachable screen cannot break launch flow. The moment it is wired in every app start depends on it.
ConvoyApiClient NOT called from any existing ViewModel until Phase 2	All existing ViewModel code operates offline. Network calls before Wednesday risk timeouts during field test.
createRide() NOT wired to API until W-03	Existing ride creation is proven working. Do not touch it until Wednesday confirms V2.4 solid.
Radio write engine NOT touched at all in V3.0	ConvoyProfileBuilder, InstallProfileUseCase, channel write — V3.0 only changes where ride JSON originates, not how it is applied.
Google HYB tiles replaced before any other task Monday	Legal risk. Must be resolved before any other V3.0 work begins.
build.gradle Sign-In dependency safe to add in Phase 1	Library addition has zero runtime impact if nothing calls it.
AndroidManifest deep link safe in Phase 1 if handler no-ops when user_id missing	Deep link that silently queues when prerequisites missing cannot break existing app.
IS_STANDALONE_BUILD flag controls Google Play vs beta APK separation	One flag in ConvoyConfig.kt. Beta APK = true, Google Play build = false. No duplicate codebases.

10. Environment Reference

ITEM	VALUE
Branch	feature/convoy-event-ride
EC2 IP	34.224.89.217 — convoy-api-server
EC2 SSH	ssh -i ~/.ssh/convoy-api-key-2.pem ec2-user@34.224.89.217
RDS Endpoint	convoy-tracker-db.cudtjxrtdbql.us-east-1.rds.amazonaws.com
RDS Database	convoy_tracker / convoy_admin
API Base URL	http://34.224.89.217/
KML Upload Path	/var/www/html/uploads/kml/{route_id}.kml
Schema SQL File	create_schema_v3_final.sql
Website	http://grouptrack.org — LIVE
Device 1	8624SBCEDF00001789 — primary test device
Device 2	24039703201775
Key pair	~/.ssh/convoy-api-key-2.pem
Google Play Services	com.google.android.gms:play-services-auth:21.x.x
Target package name	com.grouptrack.android (rename before Google Play submission)